

1. Which among the following is not expected to undergo self-aldol condensation?

I. 2-phenylacetaldehyde; II.

Trichloroacetaldehyde; III. Benzaldehyde; IV.

Acetaldehyde

a. II and III only

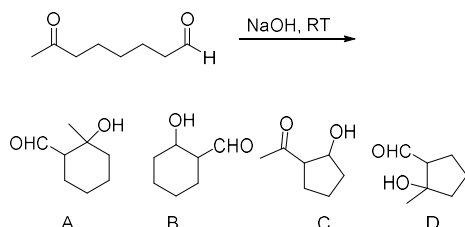
b. I, II and III only

c. II, III and IV only

d. III and IV only

2. Determine the major product of the following intramolecular aldol reaction.

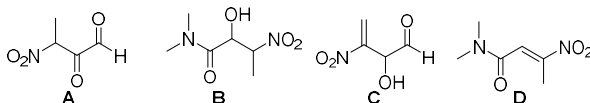
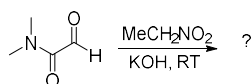
(1 mark added to all)



3. Nylon-6,6 is prepared using:

(a) adipoyl chloride, hexane-1,6-diamine, Sodium hydroxide

(b) adipoyl chloride, ethylenediamine,



Sodium hydroxide

(c) adipic acid, hexamethylenediamine, Sodium hydroxide

(d) adipic acid, ethylenediamine, Sodium hydroxide

4. Find the correct stereoisomeric relation of the following aldol products:

(1 mark added to all)

a) enantiomers

b) diastereomers

c) Homomers

d) Epimers

5. Which of the following can act as Lewis Acids? NF_3 ; SiF_4 ; BF_3 ; PF_3

a) BF_3 , PF_3 and NF_3 only

b) BF_3 , SiF_4 and PF_3 only

c) PF_3 , NF_3 and SiF_4 only

d) BF_3 , SiF_4 and NF_3 only

6. pKa of pure water is:

a) 7

b) 10^{-7}

c) 10-14

d) 15.6

7. Which combination of carbonyl compounds gives phenyl vinyl ketone by an aldol condensation?

a) Acetophenone and Acetone

b) Acetone and Benzaldehyde

c) Acetophenone and Formaldehyde

d) Formaldehyde and Acetone

8. Which among the following polymers contain only glucose unit?

a) Sucrose

b) Maltose

c) Lactose

d) none of them

9. Chose the correct product for the following nitro-aldol reaction. (option B is correct)

10. Cross Aldol reaction between cyclohexanone and acetaldehyde gives:

a) 4 stereoisomers

b) 8 stereoisomers

c) 2 stereoisomers

d) none of the above

11. Which of the following is not a Double Salt?

- b) $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$
- c) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
- d) $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
- e) $(\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O})$

12. Consider the violet-colored compound, $[\text{Cr}(\text{OH}_2)_6]\text{Cl}_3$ and the yellow compound, $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$. Which of the following statements is false?

- (a) Both chromium metal ions are paramagnetic with 3 unpaired electrons.
- (b) Δ_{oct} for $[\text{Cr}(\text{NH}_3)_6]^{3+}$ is calculated directly from the energy of yellow light.
- (c) Δ_{oct} for $[\text{Cr}(\text{OH}_2)_6]^{3+}$ is less than Δ_{oct} for $[\text{Cr}(\text{NH}_3)_6]^{3+}$.
- (d) The two complexes absorb their complementary colors.

13. Which of the following is a Weak field ligand?

- a) CN^-
- b) NH_3
- c) CO
- d) H_2O

14. Which of the following statement is wrong about a double salt

- a) Found in solid state only
- b) Metal ions have normal valency
- c) Are completely ionized in aqueous solution
- d) Always exhibit intense color

15. Arrange the following ligands in the increasing order of field strength. Ligand list: NO_3^- , NO_2^- , NH_3 , OH^-

- a) $\text{NO}_3^- < \text{NO}_2^- < \text{NH}_3 < \text{OH}^-$
- b) $\text{NO}_3^- < \text{NO}_2^- < \text{NH}_3 < \text{OH}^-$
- c) $\text{NO}_3^- < \text{OH}^- < \text{NO}_2^- < \text{NH}_3$
- d) $\text{NO}_3^- < \text{OH}^- < \text{NH}_3 < \text{NO}_2^-$

16. Which among the following is not a condensation polymer?

- a) Polyacrylonitrile
- b) Cellulose
- c) Polycarbonate
- d) Bakelite

17. For the $[\text{Fe}(\text{CN})_6]^{3-}(\text{aq}) + e = [\text{Fe}(\text{CN})_6]^{4-}(\text{aq})$ reaction, if the scan rate increases in cyclic

voltammetry, the peak current magnitudes for the anodic and cathodic sweeps

- a) Remain constant
- b) Decrease
- c) Increase

18. Arrange the followings in the decreasing order of magnetic moment.

(1) Co^{3+} (octahedral complex with a strong field ligand), (2) Co^{3+} (octahedral complex with a weak field ligand), (3) Co^{2+} (tetrahedral complex) (4) Co^{2+} (square planar complex)

- a) $1 > 2 > 3 > 4$
- b) $2 > 3 > 4 > 1$
- c) $3 > 2 > 4 > 1$
- d) $2 > 4 > 3 > 1$

19. Arrange the following complexes in the decreasing order of crystal field splitting.

i). $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$; ii) $[\text{Rh}(\text{H}_2\text{O})_6]^{3+}$; iii) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$; iv) $[\text{Rh}(\text{CN})_6]^{3-}$

- a) $i > iii > iv > ii$
- b) $iv > ii > iii > i$
- c) $iii > iv > ii > i$
- d) $iv > iii > ii > i$

20. Mohr salt is a source of

- a) K^+ ions
- b) Al^{+3} ions
- c) Na^+ ions
- d) Fe^{+2} ions

21. The correct path of light in a spectrofluorometer is

- a) Source lamp $\rightarrow$ excitation monochromator $\rightarrow$ sample holder $\rightarrow$ emission monochromator $\rightarrow$ Detector
- b) Source lamp $\rightarrow$ emission monochromator $\rightarrow$ sample holder $\rightarrow$ excitation monochromator $\rightarrow$ Detector
- c) Detector $\rightarrow$ emission monochromator $\rightarrow$ sample holder $\rightarrow$ excitation monochromator $\rightarrow$ source lamp
- d) Source lamp $\rightarrow$ Sample holder $\rightarrow$ excitation monochromator $\rightarrow$ emission monochromator $\rightarrow$ Detector

22. In a spectrophotometric cell of 2.0 cm path

length, the solution of a substance shows the absorbance value of 1.0. If the molar

absorptivity of the compound is $2 \times 10^4 \text{ L mol}^{-1} \text{ cm}^{-1}$, calculate the concentration of the substance in solution. What is the concentration of the substance in that solution?

- a) $2.5 \times 10^{-5} \text{ mol L}^{-1}$
- b) $2.5 \times 10^{-5} \text{ mol L}$
- c) $1.0 \times 10^4 \text{ mol L}^{-1}$
- d) $1.0 \times 10^4 \text{ mol L}$

23. Which of the following is NOT TRUE about micelle formation?

- a) ΔS of micelle formation is positive
- b) The hydrophobic part lies towards the interior of micelle.
- c) The hydrophilic part lies towards the surface of micelle.
- d) ΔS of micelle formation is negative

24. The formation of micelle takes places above a particular temperature, called as:

- a) CMC
- b) Kraft Temperature
- c) Curie Temperature
- d) Critical Temperature

25. Which of the following statements is correct regarding the conductivity of solutions of electrolytes?

- a) It is independent of the size of the ions
- b) It is independent of the viscosity of the solution
- c) It depends on the solvation of ions present in solution
- d) It decreases with temperature

26. What are the selection rules for FTIR absorption?

- a) Absorption only occurs for symmetrical molecules.
- b) Absorption only occurs with a dipole change.
- c) Absorption requires an odd vibrational quantum number.
- d) Absorption requires promotion to a new quantum level.

27. What type of voltage waveform used for cyclic voltammetry experiments

- a) Sine
- b) Square
- c) Triangular
- d) Sawtooth

28. A light square wireframe each side of which is 10 cm long hangs vertically in the water with one side just touching the water surface. Find the additional force necessary to pull the frame clear of the water ($T=0.074 \text{ N/m}$)

- a) 0.0074 N
- b) 1.48 N
- c) 0.0148 N
- d) 0.74N

29. Which of the following statements is correct with respect to electrolytic solutions?

- a) Its conductance increases with dilution.
- b) Its conductance decreases with dilution.
- c) Its conductivity increases with dilution.
- d) Its equivalent conductance decreases with dilution.

30. 10 . How the stoke shift will be calculated

(where $X = \lambda_{abs}^{max} =$ absorption (in wavelength), $Y = \lambda_{emi}^{max} =$ emission (in wavelength))

- a) $X+Y$
- b) $X - Y$
- c) $\frac{XY}{X+Y}$
- d) None of the above